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H2

Started on	Tuesday, 25 August 2026, 5:09 PM
State	Finished
Completed on	Tuesday, 25 August 2026, 5:10 PM
Time taken	34 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:

The first line contains the no of elements in the list-n

The next n lines contain the elements.

Output:

Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```

1  #include <stdio.h>
2
3  void swap(int *a, int *b) {
4      int temp = *a;
5      *a = *b;
6      *b = temp;
7  }
8
9  int partition(int arr[], int low, int high) {
10     int pivot = arr[high];
11     int i = (low - 1);
12
13     for (int j = low; j < high; j++) {
14         if (arr[j] < pivot) {
15             i++;
16             swap(&arr[i], &arr[j]);
17         }
18     }
19     swap(&arr[i + 1], &arr[high]);
20     return (i + 1);
21 }
22
23 void quickSort(int arr[], int low, int high) {
24     if (low < high) {
25         int pi = partition(arr, low, high);
26
27         quickSort(arr, low, pi - 1);
28         quickSort(arr, pi + 1, high);
29     }
30 }
31
32 int main() {
33     int n;
34     if (scanf("%d", &n) != 1) return 0;
35
36     int arr[n];
37     for (int i = 0; i < n; i++) {
38         scanf("%d", &arr[i]);
39     }
40
41     quickSort(arr, 0, n - 1);
42
43     for (int i = 0; i < n; i++) {
44         printf("%d", arr[i]);
45         if (i < n - 1) {
46             printf(" ");
47         }
48     }
49     printf("\n");
50
51     return 0;
52 }
```

	Input	Expected	Got	
✓	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✓
✓	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✓
✓	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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